



Project Title:

Transjakarta: Jakarta Public Bus Interconnectivity and Coverage

Introduction

A precise and succinct description of what findings and messages you wanted your narrative visualisation to convey, and who your intended audience is.

Jakarta, the Capital Region of Indonesia, currently has The longest Bus Rapid Transit (BRT) system in the world (Seasia Stats, 2024). Despite that, the projection of Jakarta becoming the largest metropolitan area by 2030 (Jauregui-Fung, 2022) produces more pressing issues of interconnectivity and accessibility for its people. Due to that, data-driven insight is crucial to support the Transjakarta's executives and Government bodies to decide what improvement needed, particularly for the bus stops, to accommodate the potential increasing demand in the future.

We plan to analyze the historical and transaction data with the bus stops construction details to answer:

- How do bus stop usage patterns and traffic volumes affect route efficiency?
- What are the user profiles across routes, and how can this inform service improvements?
- Which bus stops require upgrades to accommodate increasing passenger demand?

We plan to provide the decision makers the insight from multi perspectives, delivering the narratives of ongoing issues and the further improvement to improve the capabilities of Transjakarta service which not only benefit for the Jakarta region in the future, but also as an effort to be example for other region or cities to build their public transport system effective and efficiently accommodate their people needs.

Design Process and Justifications

This section describes and justifies the iterative process that led from initial sketches to the final interactive narrative visualization. Each of the five design sheets (provided in the Appendix) is referenced in turn, and design decisions are grounded in theoretical principles.

1 Overview of Iterations

Design Sheet 1 (Brainstorming):

- Generated a broad list of candidate chart types: bar, line, pie, flow-map, connectivity graph, density plot.
- Clustered these into three thematic groups—User Profile, Connectivity/Traffic, Features—and eliminated overly complex or redundant options.
- Outcome: Focus on a composite layout combining a geographically anchored network (flow map overlaid on a Jakarta map), a two-series line chart (weekday vs weekend demand), and a population pyramid for inclusivity.

Design Sheet 2 (Alternative Designs 1):

- Explored interactive map + legend-based filtering vs multiview small multiples.
- Note that dynamic legend clicks support stakeholder-driven exploration but introduce cognitive overhead (Gestalt grouping vs change blindness).
- Decision: Maintain a single, prominent map to preserve spatial context and simplify interactions.

Design Sheet 3 (Alternative Designs 2):

- Tested a purely chart-centric layout (no map) with rich filtering controls on each chart.
- Found that removing geographic context reduced stakeholders' ability to locate bus stops regionally.
- Decision: Reinstate map as central, retaining chart filters only in the sidebar.

Design Sheet 4 (Alternative Designs 3):

- Refined interactions: zoomable map, tooltip on hover, time and corridor filters.
- Balanced user-driven (free exploration) vs author-driven (guided narrative pop-ups) per Segel & Heer's taxonomy (2010).
- Decision: Adopt a drill-down story structure—map first, then click to reveal detailed charts—supporting both overview and details-on-demand (Shneiderman, 1996).

Design Sheet 5 (Realisation):

- Finalized layout: left sidebar holds narrative prompts, dropdown, and two charts; right canvas holds the map + legend.
- Data pipeline: heavy wrangling in R (tidyverse/jsonlite) produces lightweight JSON for D3, decoupling processing from rendering.
- Chosen interactions: click-again to reset, hover tooltips, dropdown "All Stops" option.

2 Genre & Narrative Style

Slide-Show Metaphor: The visualization adopts a slide-show genre with layered interactivity. Each "slide" or view focuses on one insight—spatial connectivity, demand patterns, or rider demographics—while allowing stakeholders to control flow via clicks or dropdown selection. This mixed author/user-driven narrative balances clear messaging (author cues) with exploratory freedom (user zoom and filter), in line with Segel & Heer's typology of narrative visualization (2010).

3 Layout Structure

Drill-Down Story: A “map-first” layout (sheet 4) anchors the user in Jakarta’s geography, leveraging the map as a scaffold for all subsequent interactions (Munzner’s Why: explore geographical relationships). Sidebar charts update on click, enabling a top-down drill-down from city-wide overview to stop-level detail. This respects human visual capacity limits by presenting one primary view at a time and avoiding small multiples that exceed working memory.

4 Visual Variables & Colour Palette

Colour Encoding:

Corridors are colored with D3’s category10 scale for high discriminability. The semi-opaque gray legend background (50% opacity) ensures text legibility while maintaining map context (Ware, 2013). Weekend lines are rendered in vibrant steelblue, weekday lines in warm tomato orange, exploiting pre-attentive color channels to distinguish series.

Population Pyramid Colours:

Male/female bars use culturally familiar blue and pink hues, reinforcing inclusivity (sheet 5). A Viridis palette was considered for color-blind safety in future extensions..

5 Typography & Branding

All text employs Jakarta+, the city’s open-source branding typeface, ensuring cultural resonance and consistency with local government materials. Headings use 16 px bold, body text 12 px regular, establishing a clear visual hierarchy.

6 Interaction Design & Human Visual System

- Hover Tooltips:
Tooltips appear on pointer hover, presenting four key stop attributes. CSS white-space: pre-line collapses code indentation, improving readability (sheet 4).
- Click-to-Filter and Reset:
Clicking a stop triggers sidebar updates; a second click resets to “All Stops.” This leverages recognition-rather-than-recall and supports exploratory analysis without modal dialogs.

7 Theoretical Framework (Munzner’s What-Why-How)

- Why (Tasks): Discover spatial interconnectivity, compare weekday vs weekend demand, assess demographic inclusivity, and evaluate stop readiness for upgrades.
- What (Data): Geolocated network edges, time-series ridership counts, and age/sex population counts.
- How (Encoding & Interaction):
 - Map projection + colored nodes/links for spatial relationships.
 - Line chart for temporal comparison.
 - Back-to-back bar chart (pyramid) for demographic distribution.
 - Direct manipulation via clicks, dropdown, and hover tooltips.

Each encoding aligns with compatibility principles, length for quantitative data, color hue for categorical corridors, spatial position for geography and leverages Gestalt grouping (proximity, similarity) to organize elements meaningfully.

Implementation of Final Visualisation Design

5.1 Technical Implementation

The interactive narrative visualization is implemented in JavaScript and HTML, leveraging D3 v7 for all chart and map rendering, along with the topojson-client library to convert TopoJSON into GeoJSON. All data preparation—including merging stop metadata, ridership time-series, and population counts—was performed in R using the tidyverse and jsonlite packages; the resulting JSON files are thus lightweight and optimized for browser consumption.

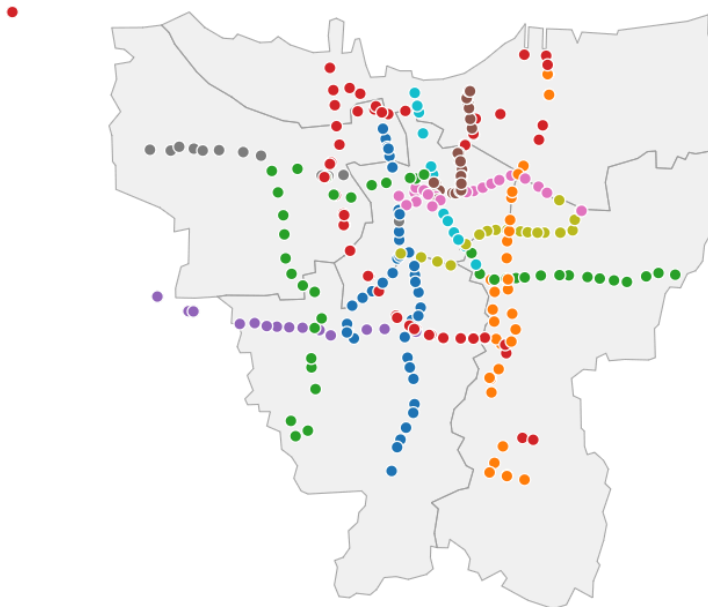
External code references include:

- D3's force-layout examples (initial prototyping)
- Mike Bostock's GeoJSON choropleth template for pan/zoom and projection setup
- TopoJSON-client documentation for the feature() import

A deviation from the originally justified design occurred in replacing the force-directed layout with a static geographic projection (via `d3.geoMercator().fitSize(...)`). This change preserves true latitude/longitude positioning, minimizes node overlap, and delivers smoother pan/zoom performance on lower-powered devices. The most technically challenging aspects were: (1) extensive R-based wrangling to produce clean JSON exports, and (2) advanced D3 coordination to overlay network links on a geographic map while maintaining responsive tooltips and chart updates.

5.2 Interactive Narrative Visualisation Implementation

1. Base Map & Network Overlay



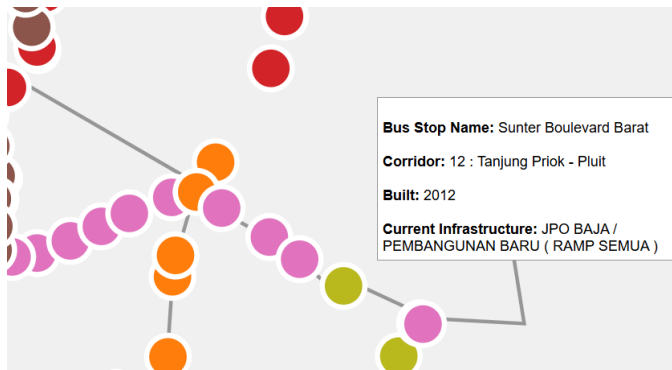
- The Jakarta outline is rendered from a simplified TopoJSON file and projected to the SVG canvas.
- Bus stops appear as color-coded circles, and corridor connections are drawn as straight lines between projected node coordinates.

2. Semi-Transparent Corridor Legend



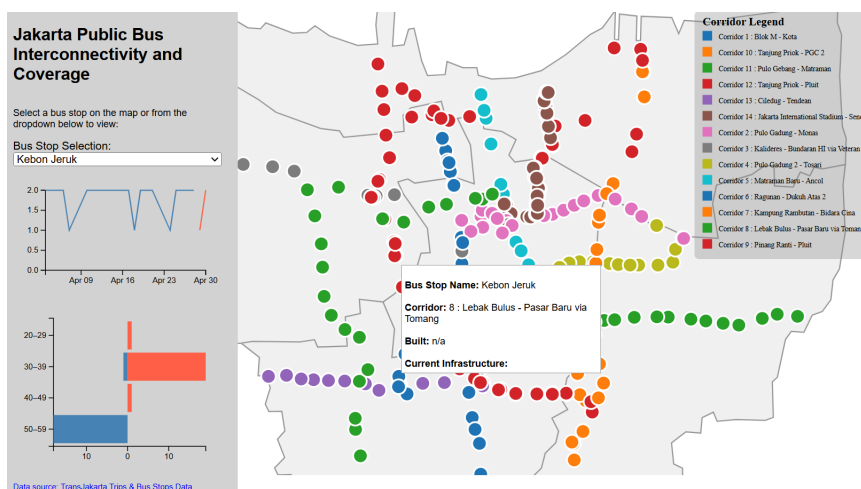
- A fixed legend box with 50% gray opacity is positioned at the top-left of the map, titled “Corridor Legend.”
- Each corridor name is paired with its D3 category10 color swatch.

3. Hover Tooltips



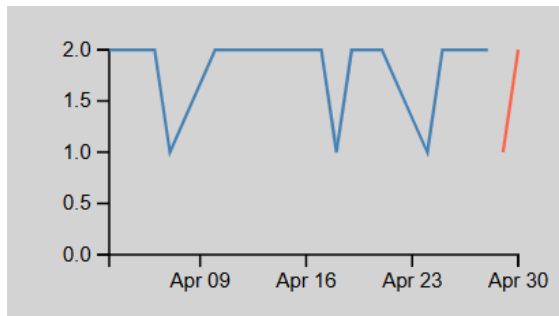
- Moving the cursor over a stop displays a tooltip with stop name, corridor, year built, and infrastructure remarks.
- CSS `white-space: pre-line` ensures neat line breaks without preserving code-level indentation.

4. Click-to-Filter & Reset Behavior



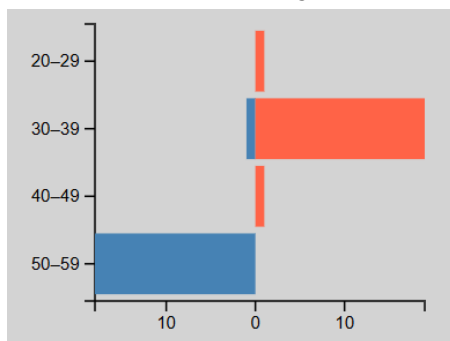
- Clicking a stop updates the sidebar charts to that stop's ridership and population data.
- A second click on the same stop (or selecting "All Stops" from the dropdown) restores overall city-wide trends.

5. Sidebar Line Chart



- A dual-series line chart contrasts weekday vs weekend ridership over time, with axes and minimal grid lines for clarity.

6. Sidebar Population Pyramid



- A back-to-back bar chart displays male and female rider counts by age group, aligned on a shared vertical axis.

5.3 Using the Implementation

1. Setup & Launch

- Place `index.html`, the four JSON data files (`bus_stop_network.json`, `bus_stop_timeseries.json`, `bus_stop_population.json`, `jakarta-simplified-topo.json`), and any supporting assets in a single directory.
- Serve via an HTTP server (e.g. `python3 -m http.server 8000`) and navigate to `http://localhost:8000/index.html`.

2. Map Interaction

- **Pan/Zoom:** Mouse wheel or pinch gesture to zoom; click-drag to pan.
- **Hover:** Tooltip appears when hovering over a stop circle.

3. Stop Selection

- **Click:** Selects a stop to update both the ridership line chart and the population pyramid.
- **Click again** (or choose “All Stops”): Resets both charts to display aggregated data for every stop.

4. Accessibility Notes

- All interactions are keyboard-accessible: Tab to focus the dropdown or circles, Enter to select.
- If JSON loads fail due to CORS, use a local HTTP server instead of `file://`.

These usage instructions ensure that any user can explore and extract insights from the visual narrative without missing key interactive features.

Conclusion

The interactive narrative visualization successfully implemented spatial, temporal, and demographic perspectives on Jakarta's bus network. By overlaying corridor connections on a geographically accurate map, it reveals patterns of inter-stop connectivity and highlights key transit hubs. The dual-series line chart contrasts weekday and weekend user trends, showing peak demand periods, while the population pyramid illustrates age and gender distributions at each stop, addressing both operational planning and inclusivity concerns. The click-to-filter and "All Stops" reset interactions provide a seamless drill-down workflow, enabling stakeholders to move fluidly between city-wide overviews and stop-level detail.

This project demonstrated the combined effort of data cleaning and wrangling in R with D3's visualization capabilities on developing accessible platform like websites. This process builds strategic thinking on how each tech stack will be used in the data visualization process and interconnected. Incorporating not only the technical knowledge, but also psychological impact and human factors, focusing on the primary audience from design process to implementation steps, make the consideration more robust and thorough. This will be highly beneficial for future works where the focus is not making the most innovative or aesthetically pleasing output, but more on the value that can be delivered to the target audience.

The quality data available is limited, the visualization cannot perform as fully as intended. Future works include expanding the scope of the project for not only main public bus, but also feeder/mini public bus, car-sized public vehicle (tuk-tuk like vehicle), to give more holistic and comprehensive and complex interactions that increase user engagement. The interconnectivity of provinces or even nation also exciting project to go through in the future, as the insight and designing decision process will be whole another level.

Bibliography

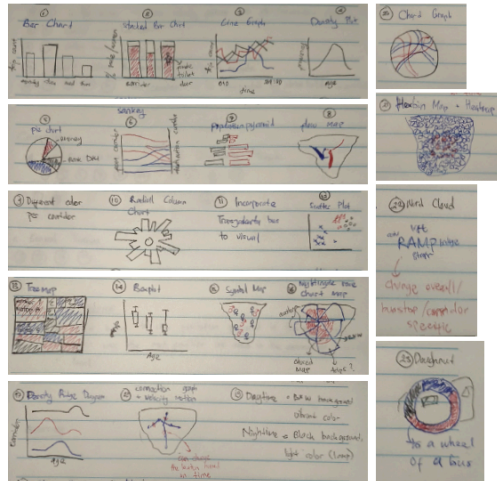
- Bertin, J. (1983). *Semiology of Graphics*. University of Wisconsin Press.
- Dikisahkan. (2023). *TransJakarta transportation transaction* [Data set]. Kaggle.
<https://www.kaggle.com/datasets/dikisahkan/transjakarta-transportation-transaction/>
- Ghapsara. (2025). *jakarta-simplified-topo.json* [TopoJSON file]. GitHub.
<https://github.com/ghapsara/indonesia-atlas/blob/master/kabupaten-kota/Jakarta/jakarta-simplified-topo.json>
- Jauregui-Fung, F. (2022). *BRT Transjakarta: Phasing in, performing and expanding a new system within a consolidated urban area: Report for the "Inclusive and sustainable smart cities in the framework of the 2030 Agenda for Sustainable Development" Project*. German Institute of Development and Sustainability (IDOS).
<https://doi.org/10.23661/r6.2022>
- Satu Data Jakarta. (2023). *Data lokasi TransJakarta DKI Jakarta* [Data set].
https://satudata.jakarta.go.id/open-data/detail?kategori=dataset&page_url=data-lokasi-transjakarta-dki-jakarta&data_no=1
- Seasia Stats. (2024, March). *The world's longest bus rapid transit (BRT) systems* [Instagram post]. Instagram.
<https://www.instagram.com/seasia.stats/p/C6h4kBRSCDy/>
- Segel, E., & Heer, J. (2010). Narrative visualization: Telling stories with data. *IEEE Transactions on Visualization and Computer Graphics*, 16(6), 1139–1148.
- Shneiderman, B. (1996). The eyes have it: A task by data type taxonomy for information visualizations. *Proceedings 1996 IEEE Symposium on Visual Languages*, 336–343.

Appendix

Five Design Sheets

Title: Transjakarta Jakarta Public Bus Interconnectivity and Coverage
Author: Muhammad Iqbal
Date: 22 May 2025
Sheet: 1
Task: Brainstorming

1. Ideas



2. Filter

6,10,14,15,13,16,20,21 are too complicated for stakeholders → *Exclude*
 1 and 2 is similar → *Bar Chart*
 8 and 10 is similar → *Flow/Connection Graph on top of Map*
 4 and 17 is similar → *Density Chart*
 5 and 23 is similar → *Pie Chart*

3. Categorize

A. User Profile: {2, 4, 5, 7, 12} → Visualize the proportion, behaviour, and characteristics of the users
B. Connectivity/Traffic: {2, 3, 8} → Shows bus traffic over time and location
C. Features: {9, 11, 18, 19} → Improve and refine the presentation and clarity of the visualizations

4. Combine and Refine

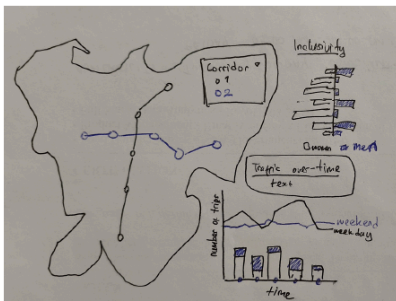
{2, 5} → Combine to compare 3+ categories and show aggregate comparisons.
 {8, 9,18,19} → Flow maps with colored corridors (also as a filter), time filters (day/night) color change the background (gray and black), and stop-level connections with relative size indicate the tap-in and tap-out.
 {2, 3,9,11} → Stacked bar and line charts to compare weekday vs weekend patterns over time, with different colors of corridor as divider and filter.

5. Question

- Does the visualizations sufficient to provide the insights to the questions?
- Is it effective enough to deliver with current combinations of visualizations?
- Does the filter is enough to give the stakeholders chance to analyze different insights?

Title: Transjakarta Jakarta Public Bus Interconnectivity and Coverage
Author: Muhammad Iqbal
Date: 22 May 2025
Sheet: 2
Task: Alternative Designs - 1

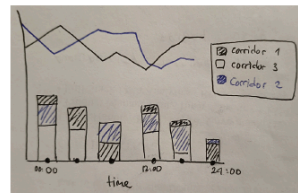
2. Layout



3. Operations

- Interactive map: Hovering over bus stops shows details (year built, accessibility, corridor, etc.).
- Legend click: Selecting a corridor in the legend filters all visualizations to highlight only that corridor (others dimmed). Click again to reset.
- Time focus: Clicking the x-axis (time) highlights specific time windows (e.g., evening rush). After 6 PM, a dark mode activates across all visuals to simulate night conditions.
- Reset: Clicking again on time/corridor restores default view.

4. Focus



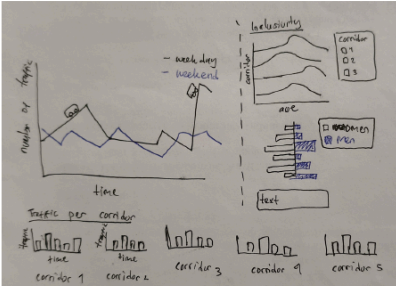
- Highlights traffic trends across corridors and time, with emphasis on comparing weekday vs weekend usage.
- The combination of line and stacked bar charts gives both detailed and aggregated insights on corridor performance.

5. Discussion

- (+) → Supports multi-perspective insights (time, location, corridor).
 - Geographic context makes corridor relevance easier to grasp.
 - Interactive features allow users to explore and extract tailored insights.
- (-) → Multiple filters (corridor + time) could increase complexity for some stakeholders.
 - May require additional onboarding or tooltip hints to guide users in using filters effectively.

Title: Transjakarta: Jakarta Public Bus Interconnectivity and Coverage
Author: Muhammad Iqbal
Date: 22 May 2025
Sheet: 3
Task: Alternative Designs - 2

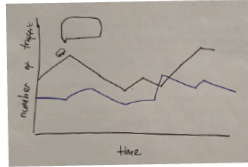
2. Layout



3. Operations

- Interactive time filtering: Users can click the bus icon on the line chart to filter data by a selected time range. Other charts will automatically update.
- Legend-based filtering: Each chart's legend can be clicked to isolate specific corridors or days. Inactive categories fade to gray (lower opacity).
- Hover interactions: Tooltips display exact values and percentages for each bar/line segment.

4. Focus



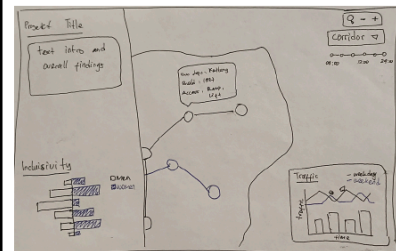
- Designed to communicate real-time trends and usage patterns to stakeholders.
- Emphasizes the comparison between weekday and weekend traffic across corridors.
- Supports planning decisions by visually breaking down trends over time and per corridor.

5. Discussion

- (+) → Rich set of visualizations to address multiple questions and insights
 - Separated bar charts by corridor enhance clarity and comparability.
 - Clear trend differentiation suitable for timely report.
- (-) → Lack of geographic map might make it difficult for some users to understand corridor locations.
 - Time filter via bus icon could be unclear without visual cues or tooltips.
 - May require onboarding or user guidance to fully utilize all filtering functions.

Title: Transjakarta: Jakarta Public Bus Interconnectivity and Coverage
Author: Muhammad Iqbal
Date: 22 May 2025
Sheet: 4
Task: Alternative Designs - 3

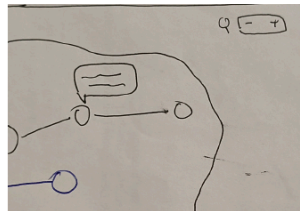
2. Layout



3. Operations

- User can zoom-in and zoom-out to make to choose between focused on each of the bus stops closer or overall glance.
- Hovering across dot in connection chart will show tooltips that provides more details of each bus stops (name, corridor, built year, accessibility)
- There are corridor (drop-down) and time range filter to filter out the data to be more focused based on the user needed, and the other categories in other chart will be grayed-out and lower opacity.

4. Focus



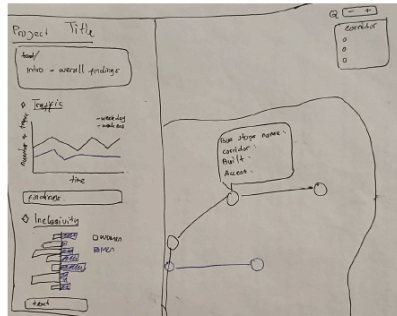
- This design emphasizes bus stop-level insights using a map-centered visualization.
- By combining location context with interactive filters (e.g., time range, corridor selection), the design supports both specific investigation and holistic overview, helping stakeholders identify stop-specific issues and broader patterns.

5. Discussion

- (+) → The centered map helps users understand where each bus stop operates, which improves spatial reasoning.
 - Zooming and filtering enable flexible exploration based on user needs, making the design more adaptable for various stakeholders.
- (-) → When charts are too small or too dense, it may become difficult for non-technical users to extract meaningful insights without guidance.

Title: Transjakarta: Jakarta Public Bus Interconnectivity and Coverage
Author: Muhammad Iqbal
Date: 22 May 2025
Sheet: 5
Task: Realisation

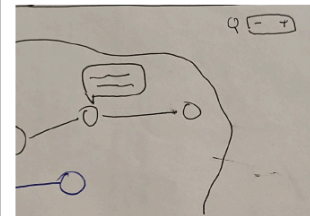
2. Layout



3. Operations

- Users can zoom in/out to view individual stops or the overall connection bus stops chart.
- Hover tooltips: Display detailed stop information (name, corridor, year built, accessibility).
- Users can filter data by corridor through the legend or by time range, with all charts and the map updating dynamically. Non-selected elements are dimmed, ensuring a consistent and focused view across the dashboard.

4. Focus



- Enables users to explore bus stop-level insights in both local detail and citywide context.
- Supports planning and policy decisions by revealing stop-specific performance and broader patterns across corridors.

5. Details

Datasets → A combined dataset consisting 2 datasets: generated datasets from kaggle with references of real historical traffic data of Transjakarta April 2023, and Bus Stops details data from Greater Jakarta Government OpenData portal of 2023. Data cleaned and wrangled using R.

Dependencies → D3 in JavaScript

Estimations → Cost :-

- Time: 29 May 2025 (Overall Initial Layout and Static Chart)
- 4 June 2025 (Filter and Interactions)
- 9 June 2025 (Finishing)